# Basics of MAC Addressing

CSE320 – Data Communications

Department of Computer Science and Engineering
School of Data & Science

# MAC Address

- Stands for Media Access Control
- Every node in the LAN is identified with MAC address
- Physical address, hardware address
- Globally unique
- Can not be altered
- Provided by manufacturer
- Represented in hexadecimal format, 48 bits long
- Example, 00:1B:44:11:3A:B7
- Separators can be: hyphen(-), colon(:) or period(.)

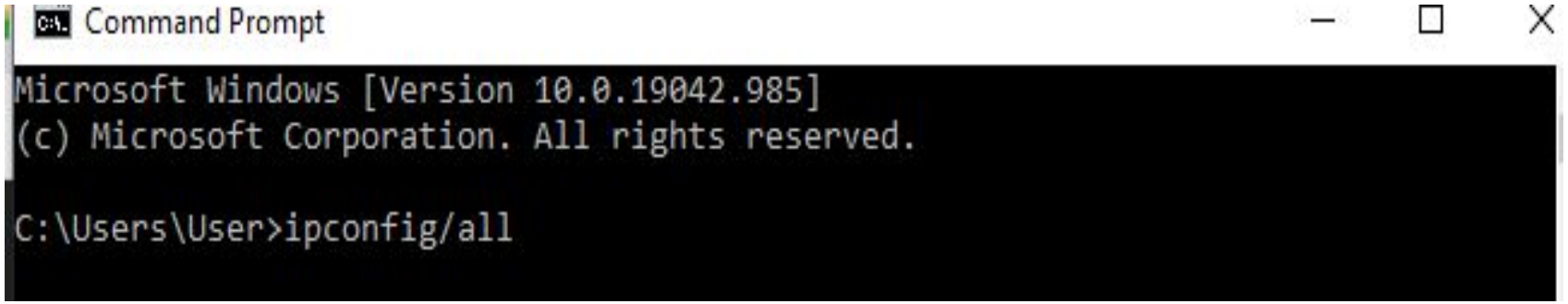
- IP Address ⟹ Location of a person
- MAC Address ⟹ Name of the person

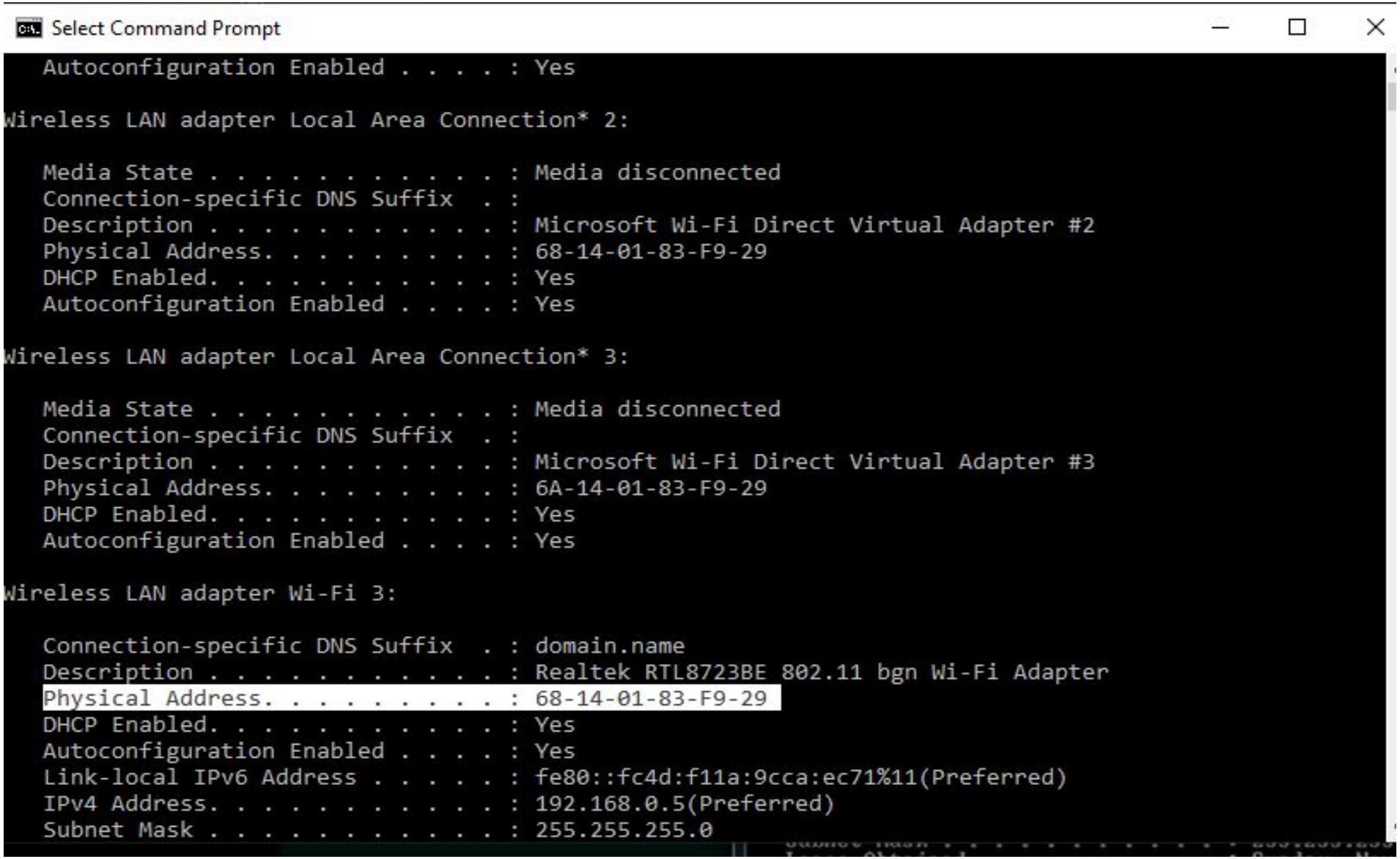
# Why do we need both IP and MAC address?

- MAC address - works on layer 2 network, identify devices within the same broadcast network
- IP address - used on layer 3, identify the devices on different networks

# How to find the MAC address of a device

1. Open command prompt, type ipconfig/all command and press enter




Select Command Prompt
Autoconfiguration Enabled . . . . : Yes
Wireless LAN adapter Local Area Connection* 2:
Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : Microsoft Wi-Fi Direct Virtual Adapter #2
Physical Address. . . . . : 68-14-01-83-F9-29
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
Wireless LAN adapter Local Area Connection* 3:
Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : Microsoft Wi-Fi Direct Virtual Adapter #3
Physical Address. . . . . : 6A-14-01-83-F9-29
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
Wireless LAN adapter Wi-Fi 3:
Connection-specific DNS Suffix . : domain.name
Description . . . . . : Realtek RTL8723BE 802.11 bgn Wi-Fi Adapter
Physical Address. . . . . : 68-14-01-83-F9-29
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::fc4d:f11a:9cca:ec71%11(Preferred)
IPv4 Address. . . . . : 192.168.0.5(Preferred)
Subnet Mask . . . . . : 255.255.255.0